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(54) **MEMORY DEVICE WITH BOTTOM ELECTRODE**

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(57) **ABSTRACT**

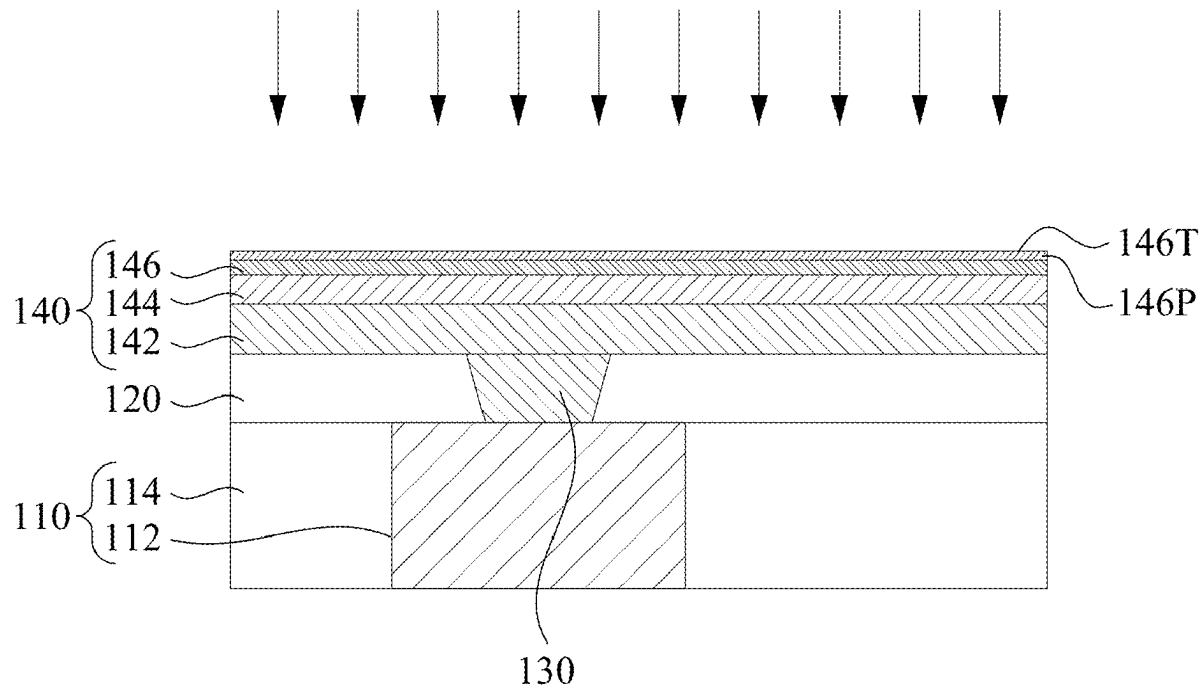
An IC structure includes an interconnect structure, a first electrode, a tantalum-containing layer, a high-k dielectric layer, and a second electrode. The first electrode is over the interconnect structure. The tantalum-containing layer is over the first electrode. The tantalum-containing layer has oxide vacancies at a top surface of the tantalum-containing layer. The high-k dielectric layer is in contact with the top surface of the tantalum-containing layer. The second electrode over the high-k dielectric layer.

(21) Appl. No.: **19/217,264**

(22) Filed: **May 23, 2025**

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(60) Continuation of application No. 18/333,145, filed on Jun. 12, 2023, now Pat. No. 12,336,442, which is a



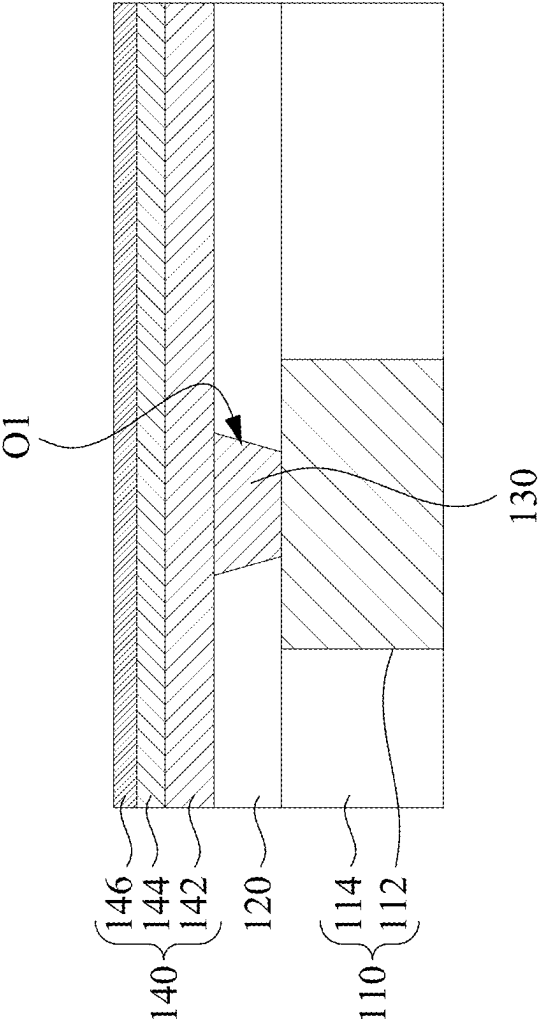


Fig. 1

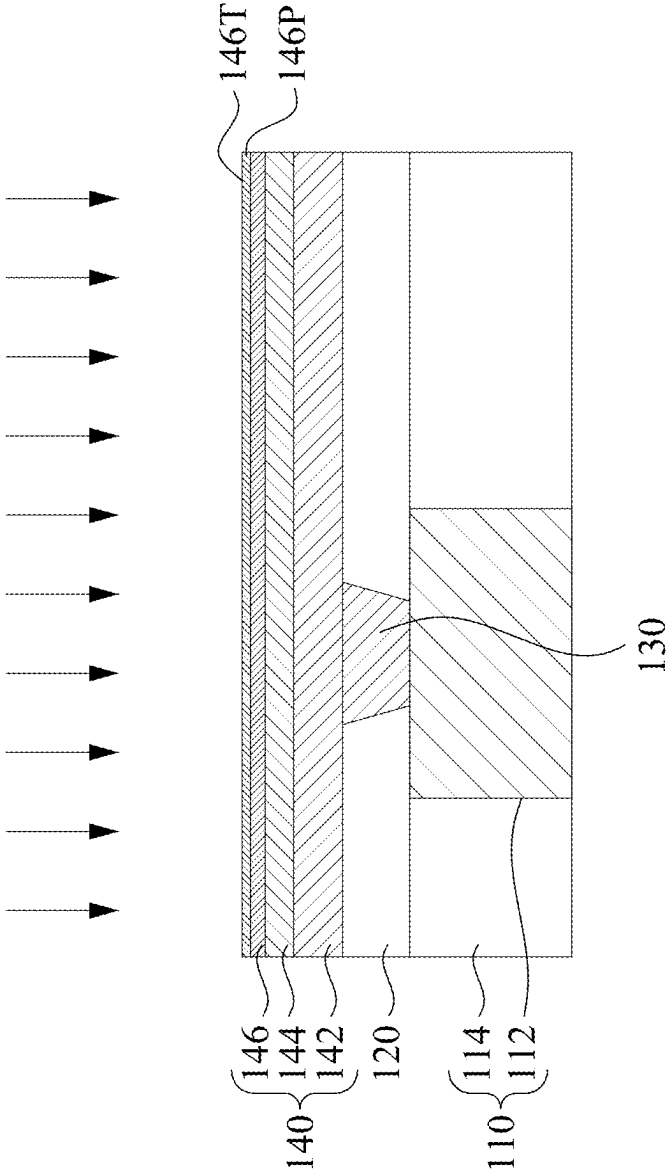


Fig. 2

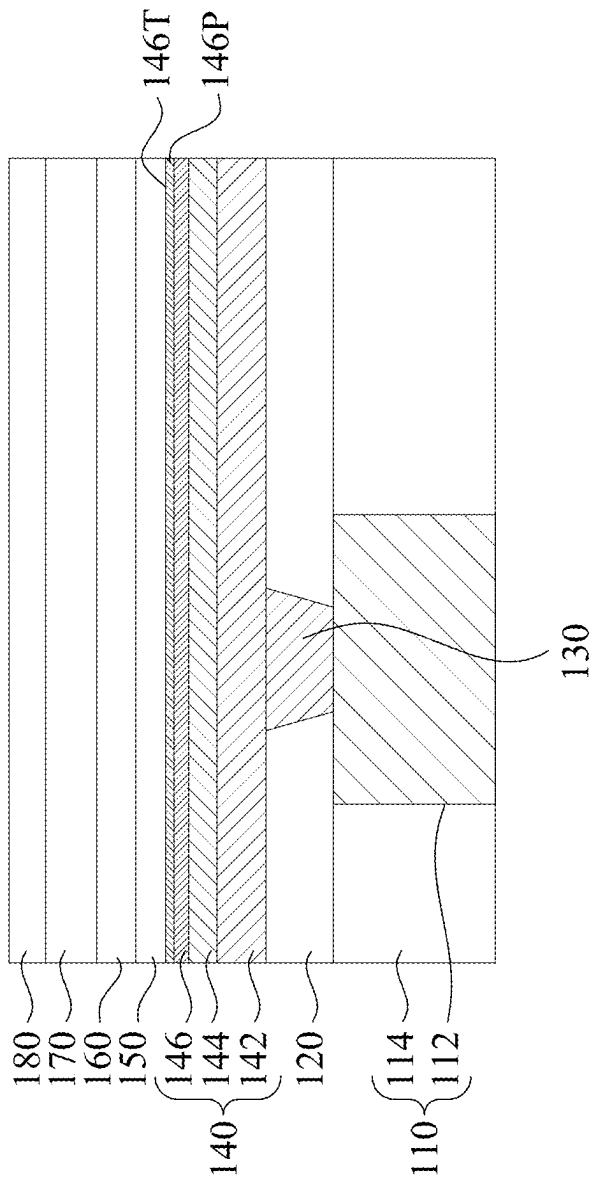


Fig. 3

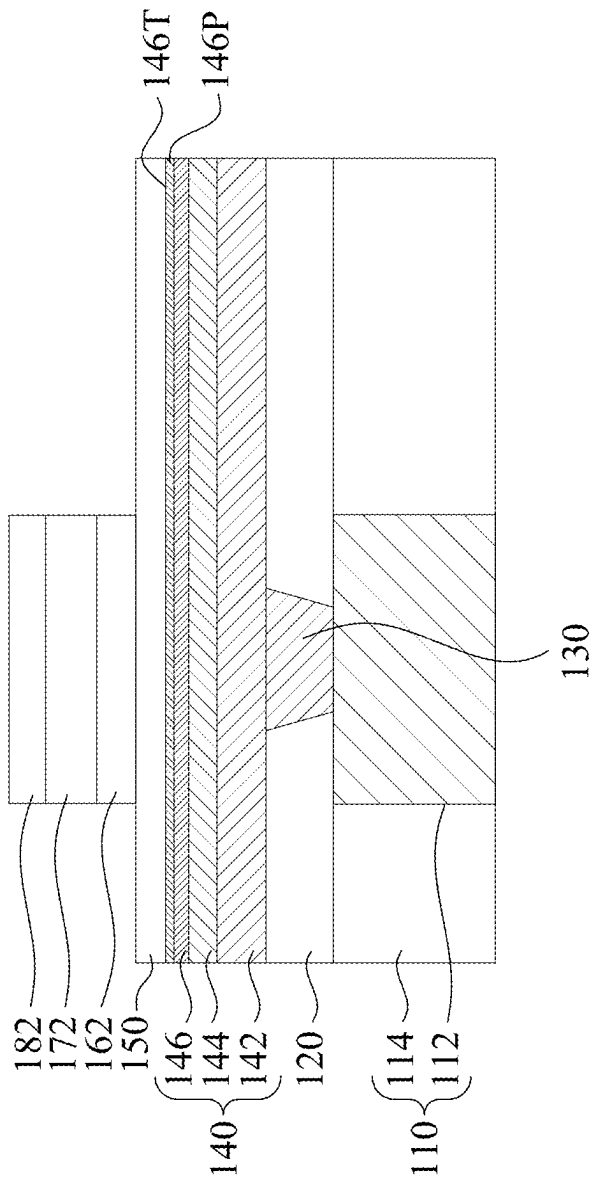


Fig. 4

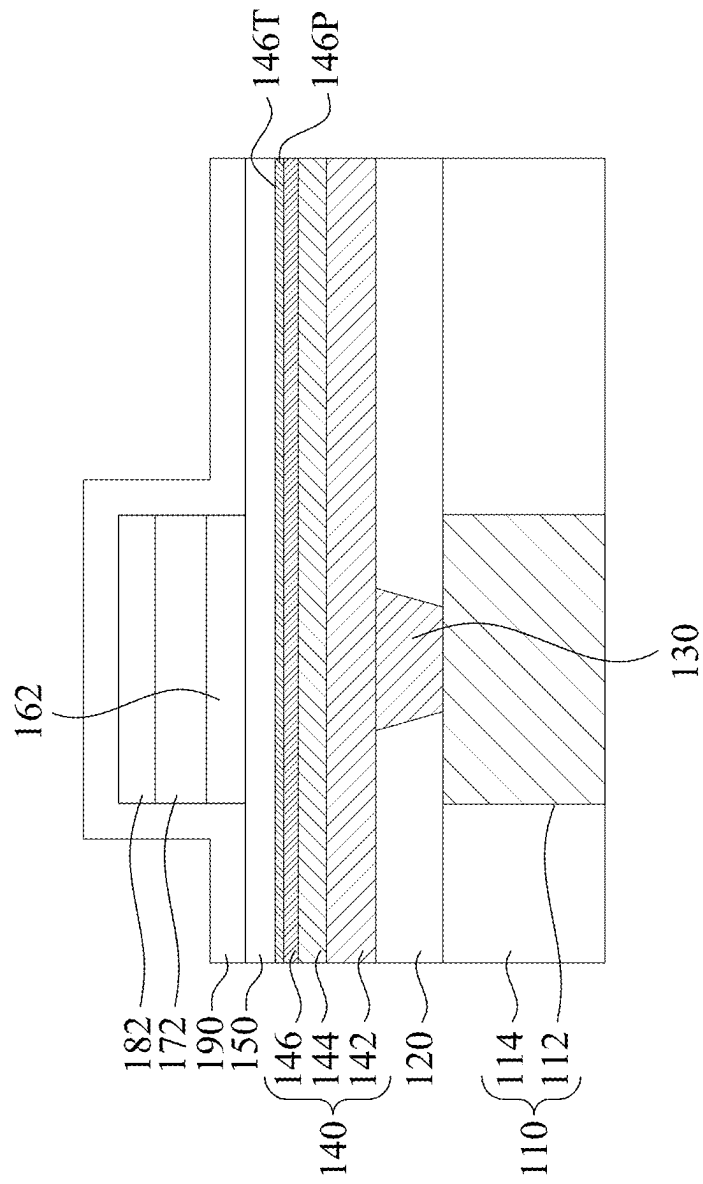


Fig. 5

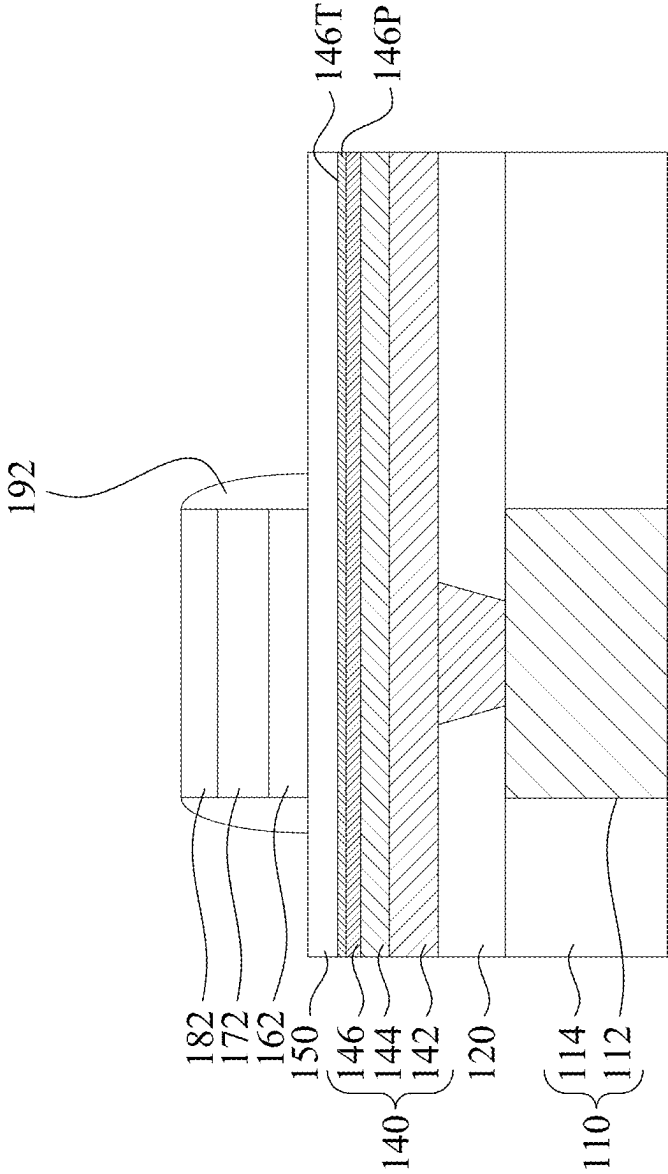


Fig. 6

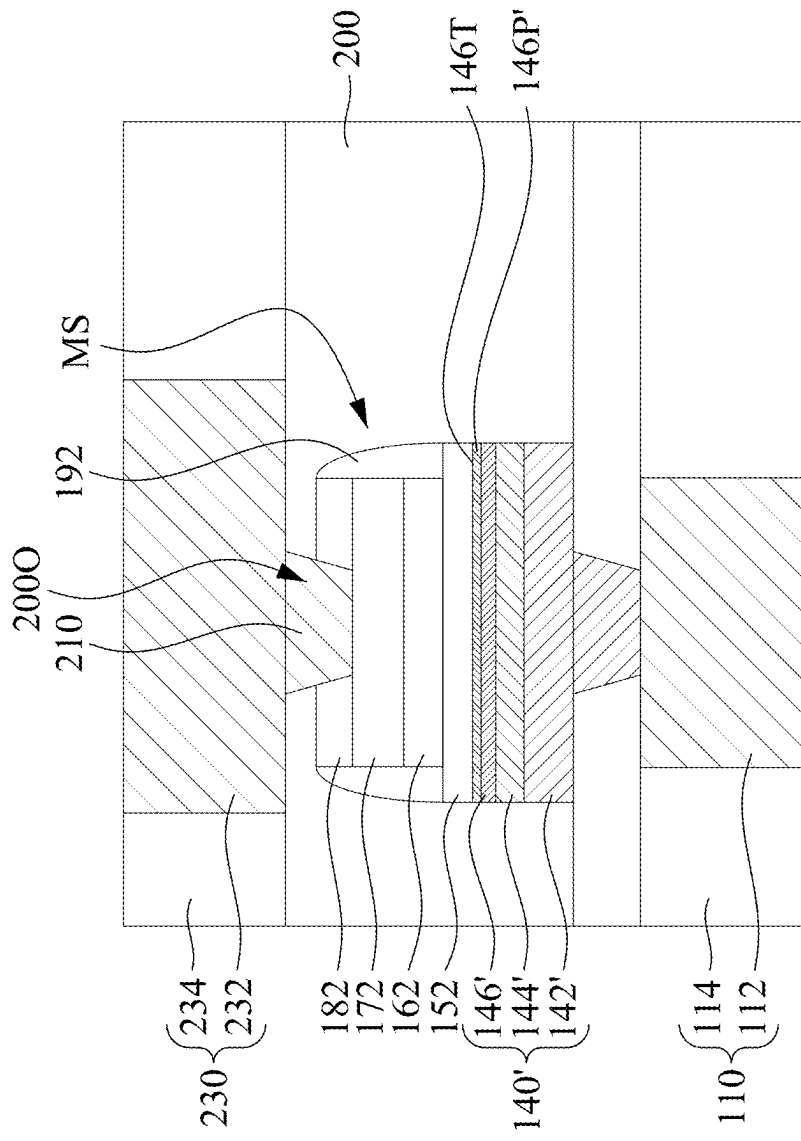


Fig. 8

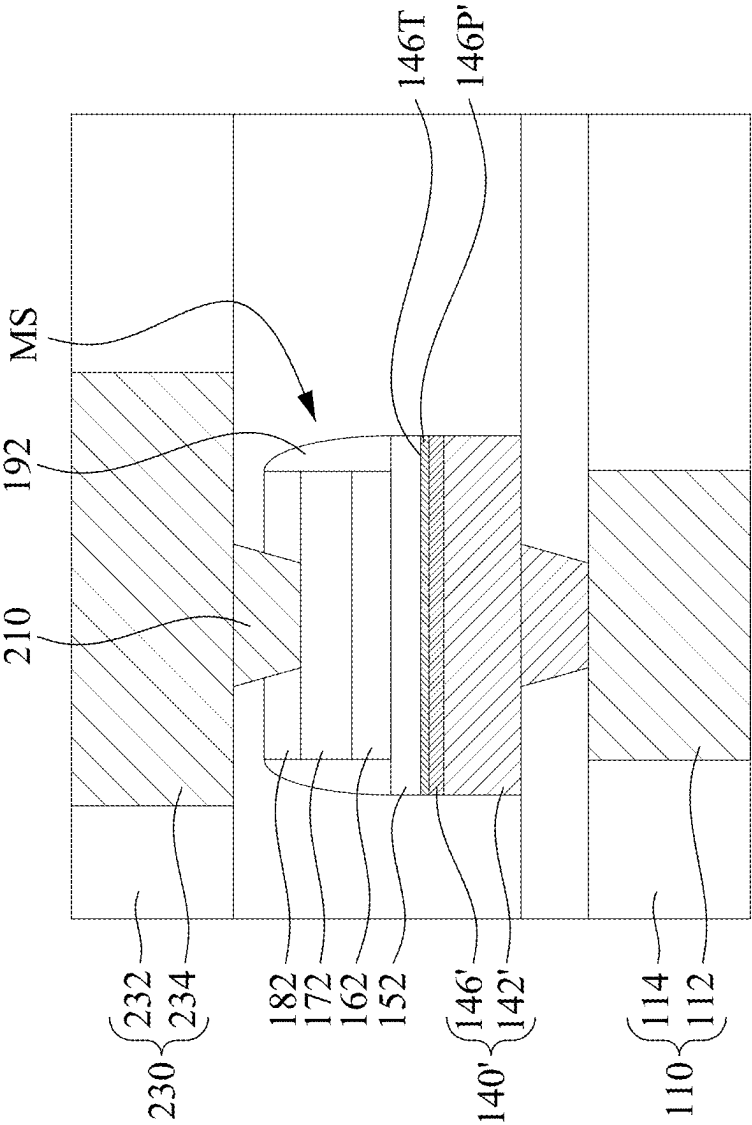


Fig. 9

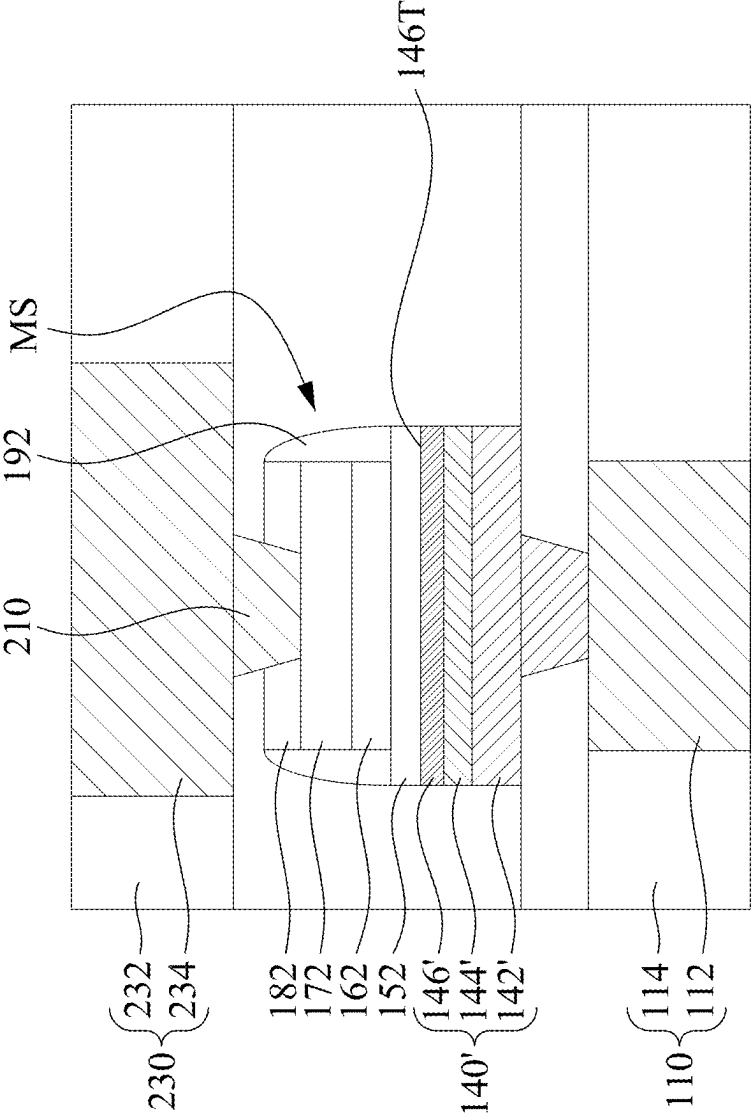


Fig. 10

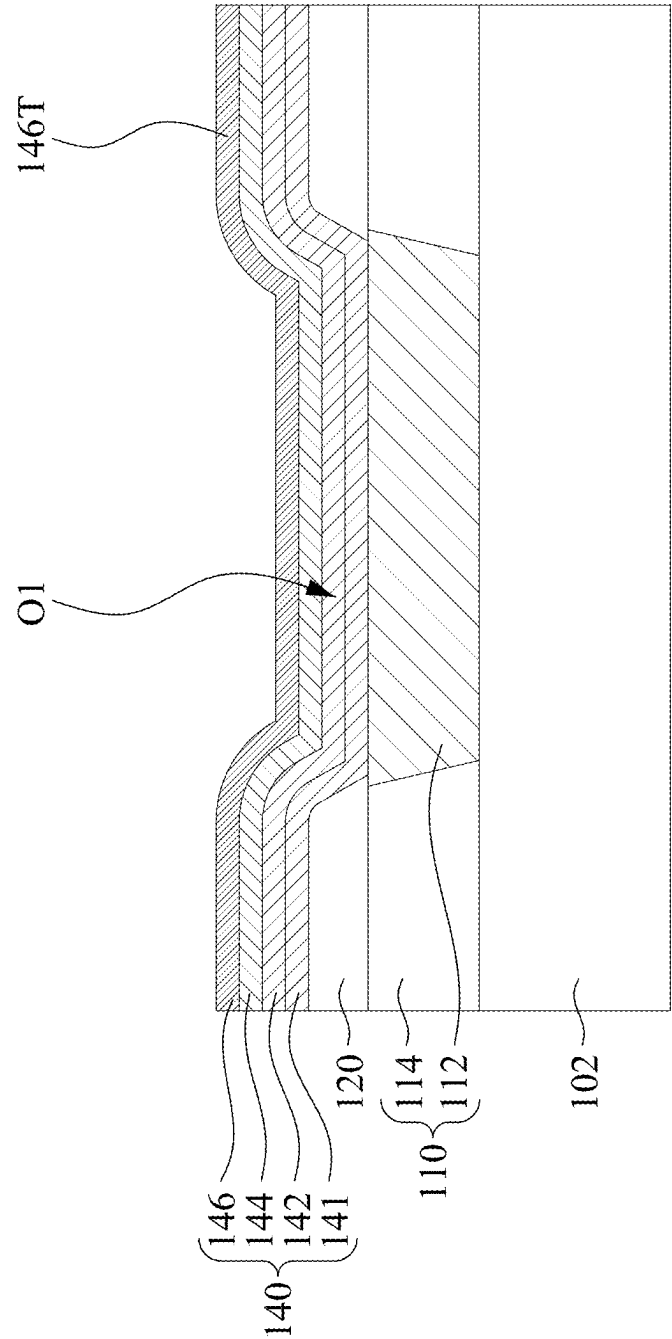


Fig. 11

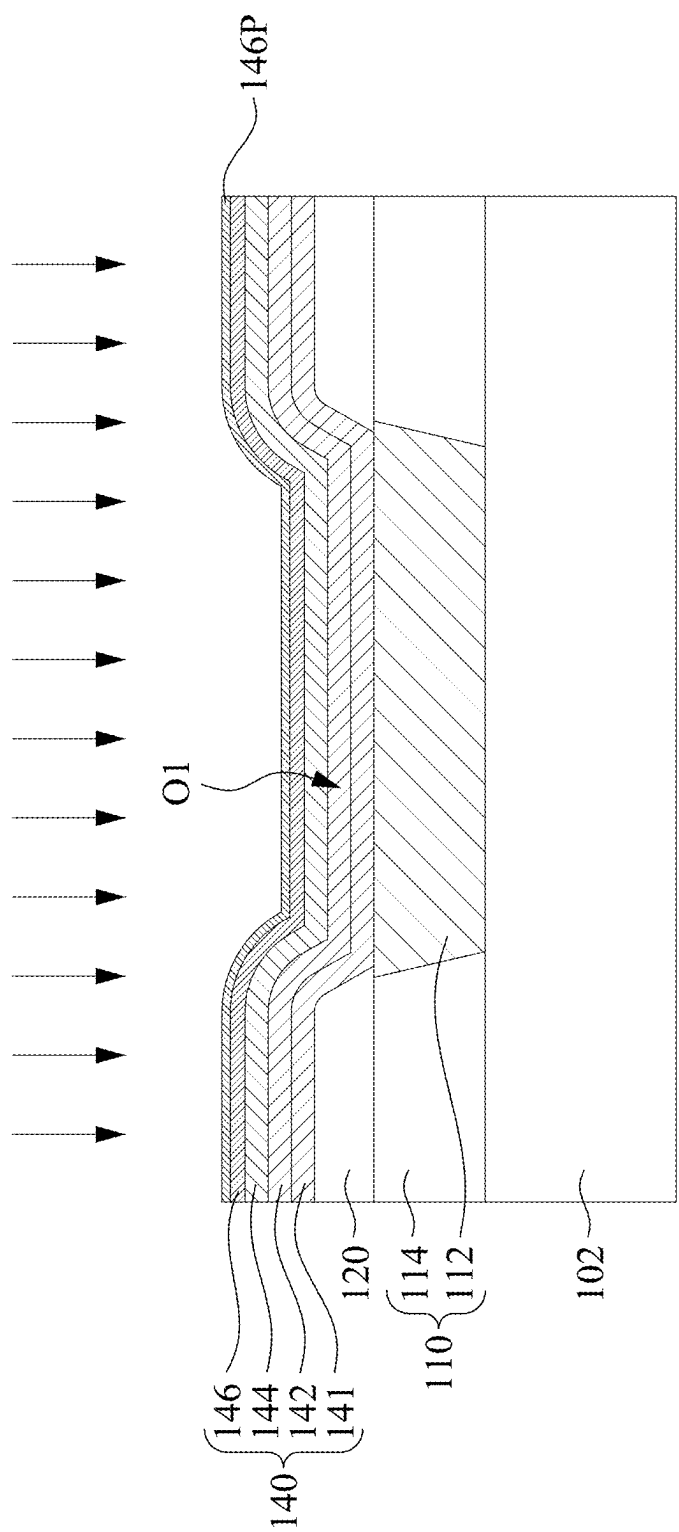


Fig. 12

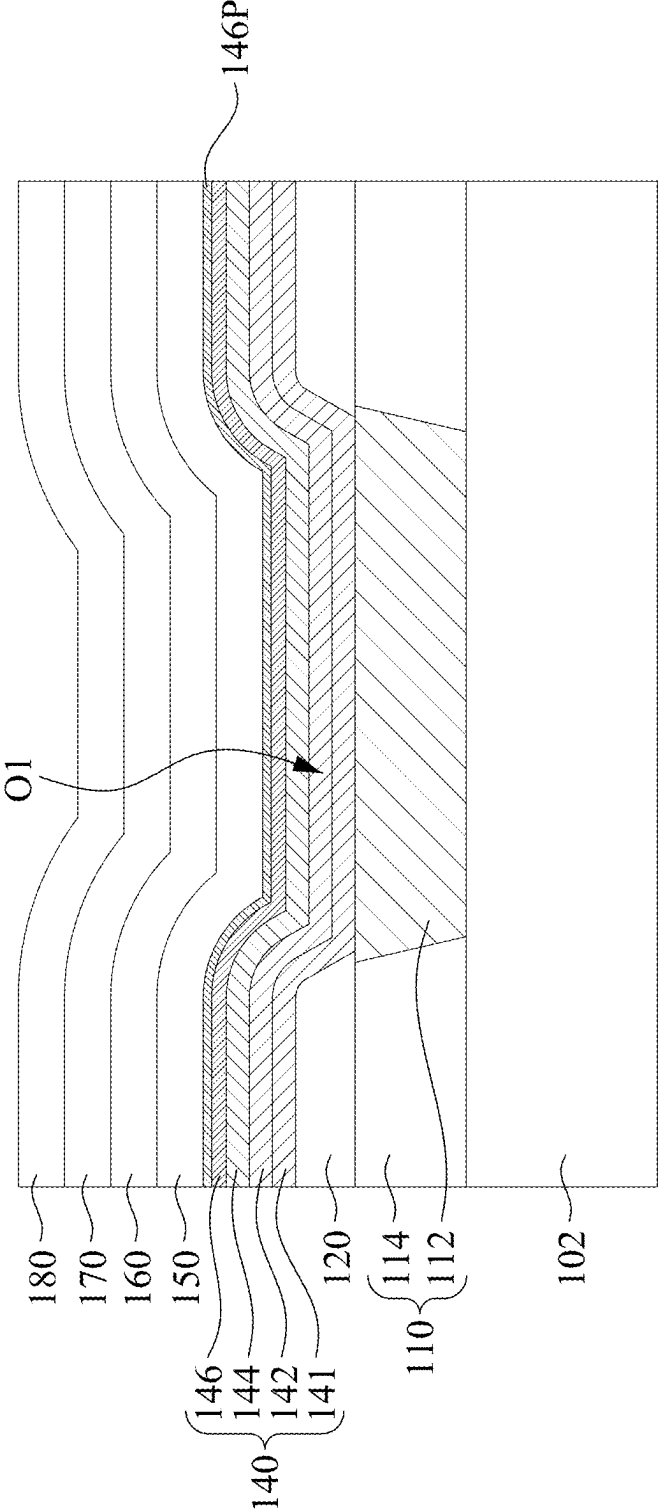


Fig. 13

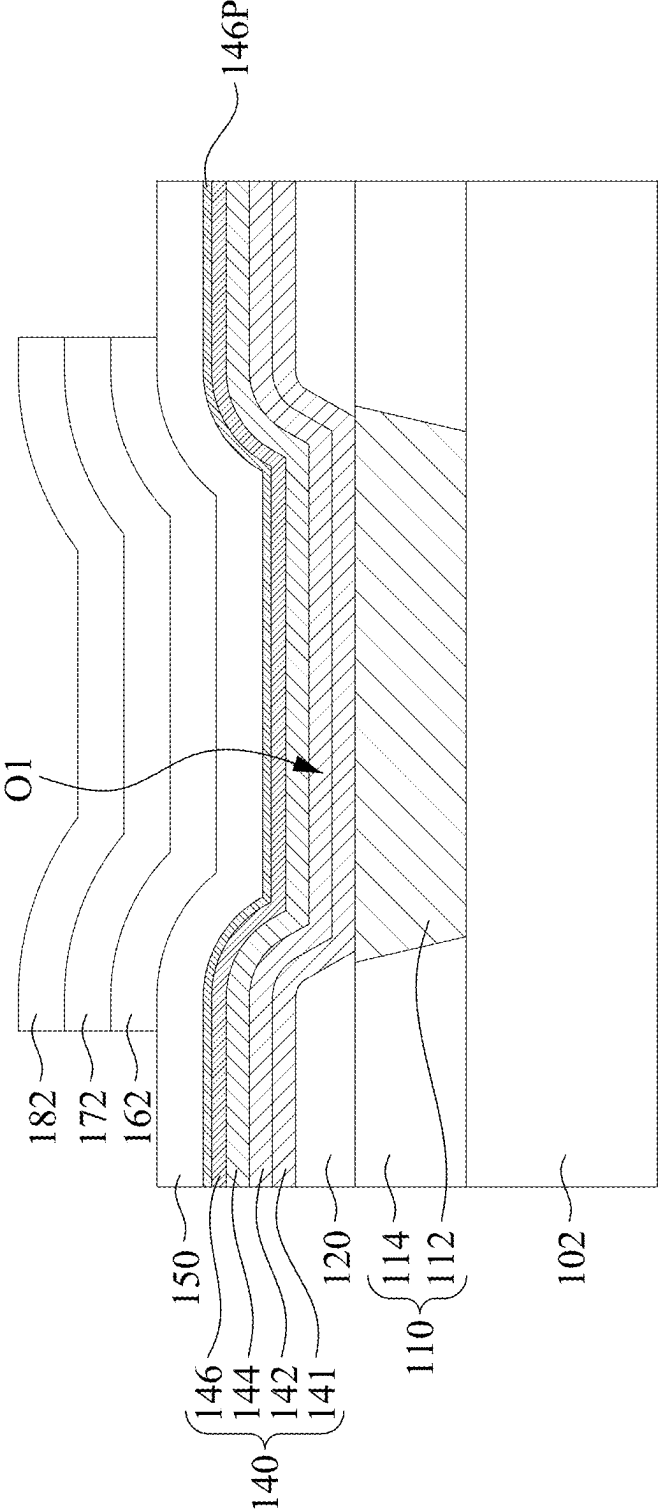


Fig. 14

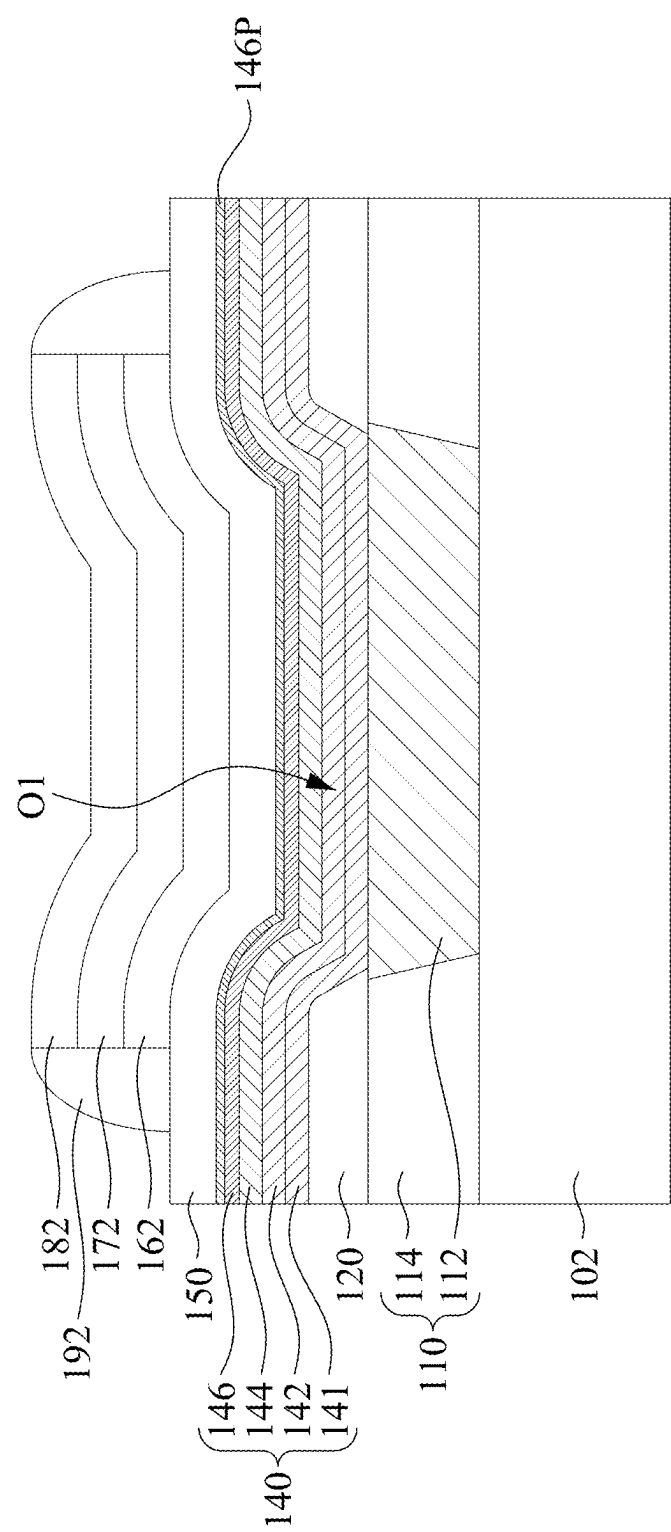


Fig. 15

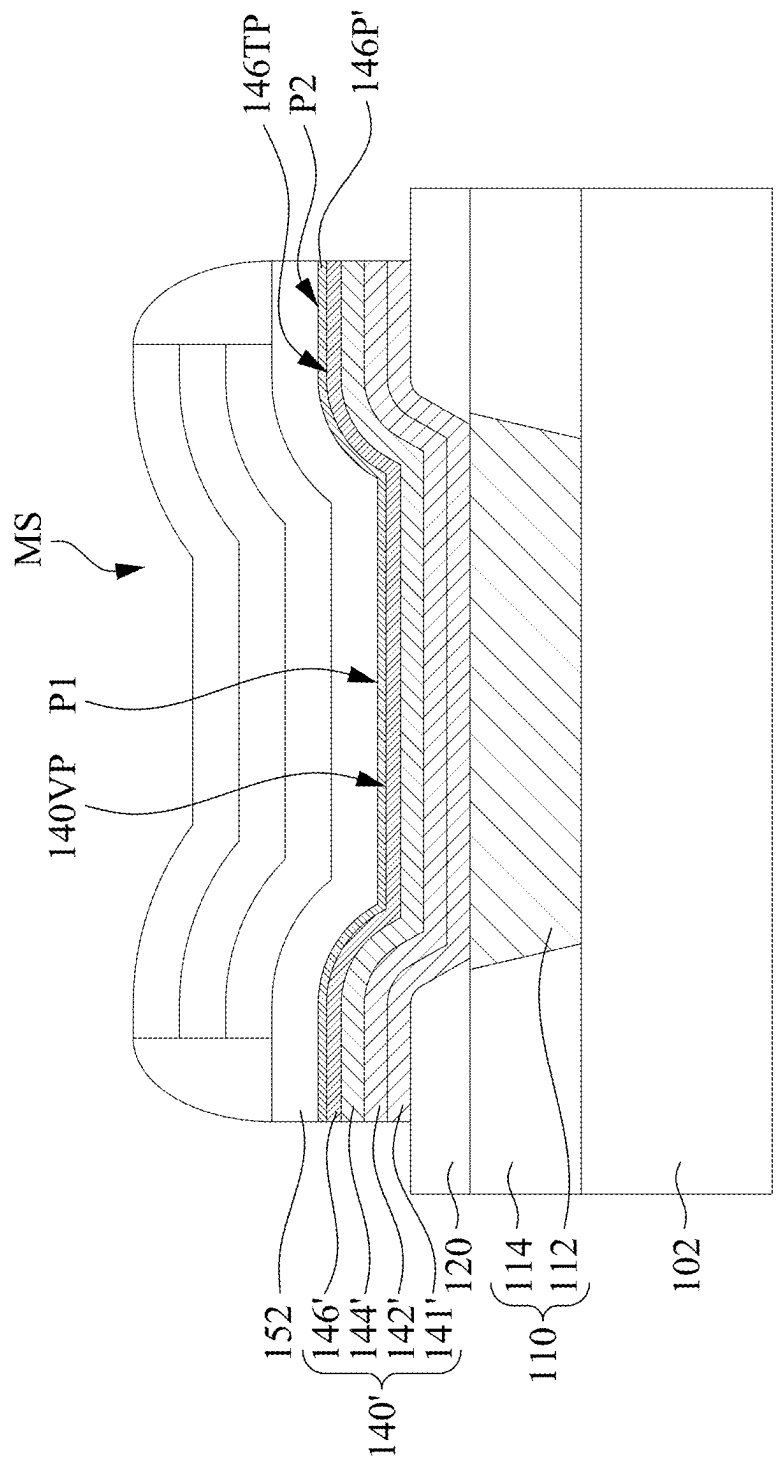


Fig. 16

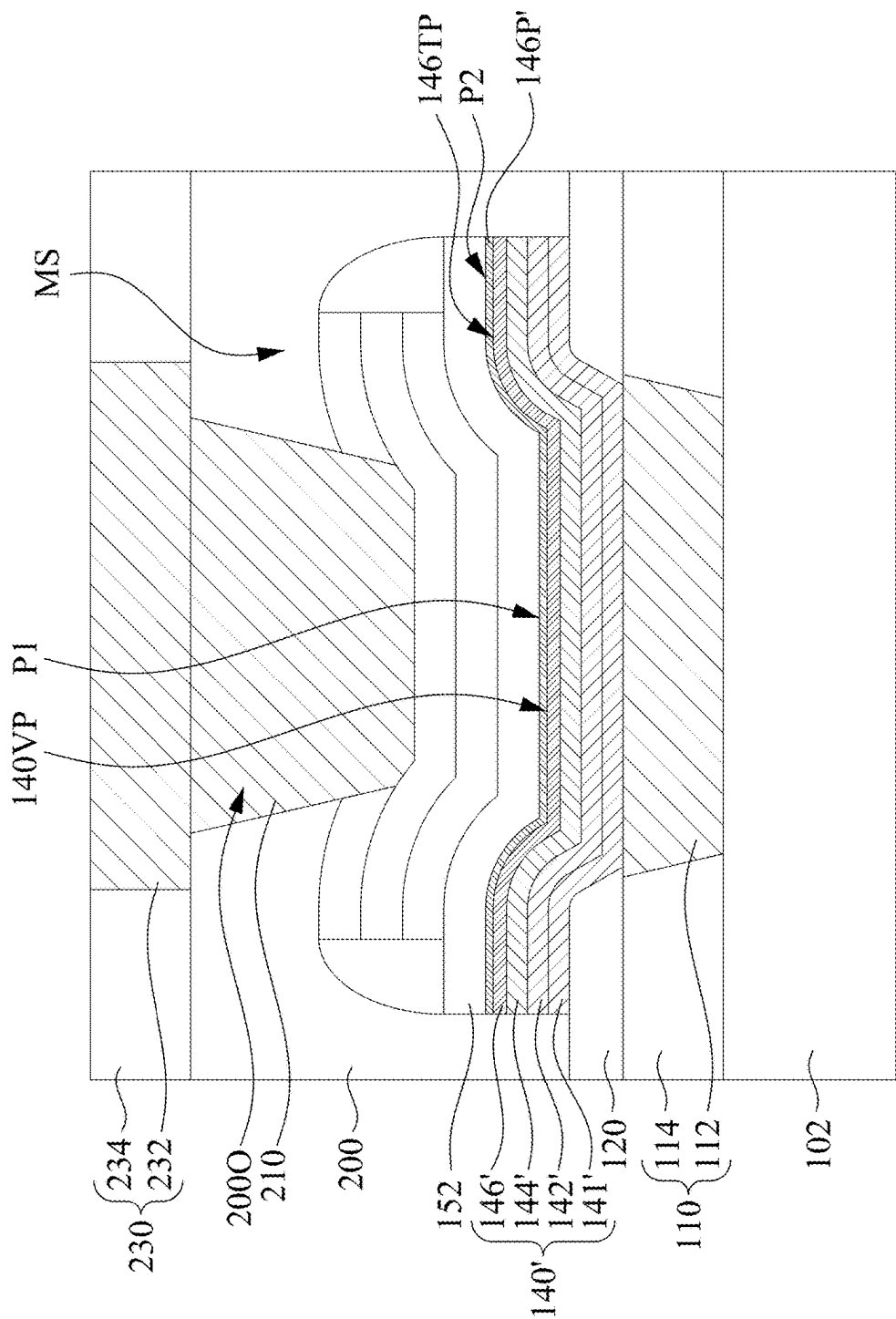


Fig. 17

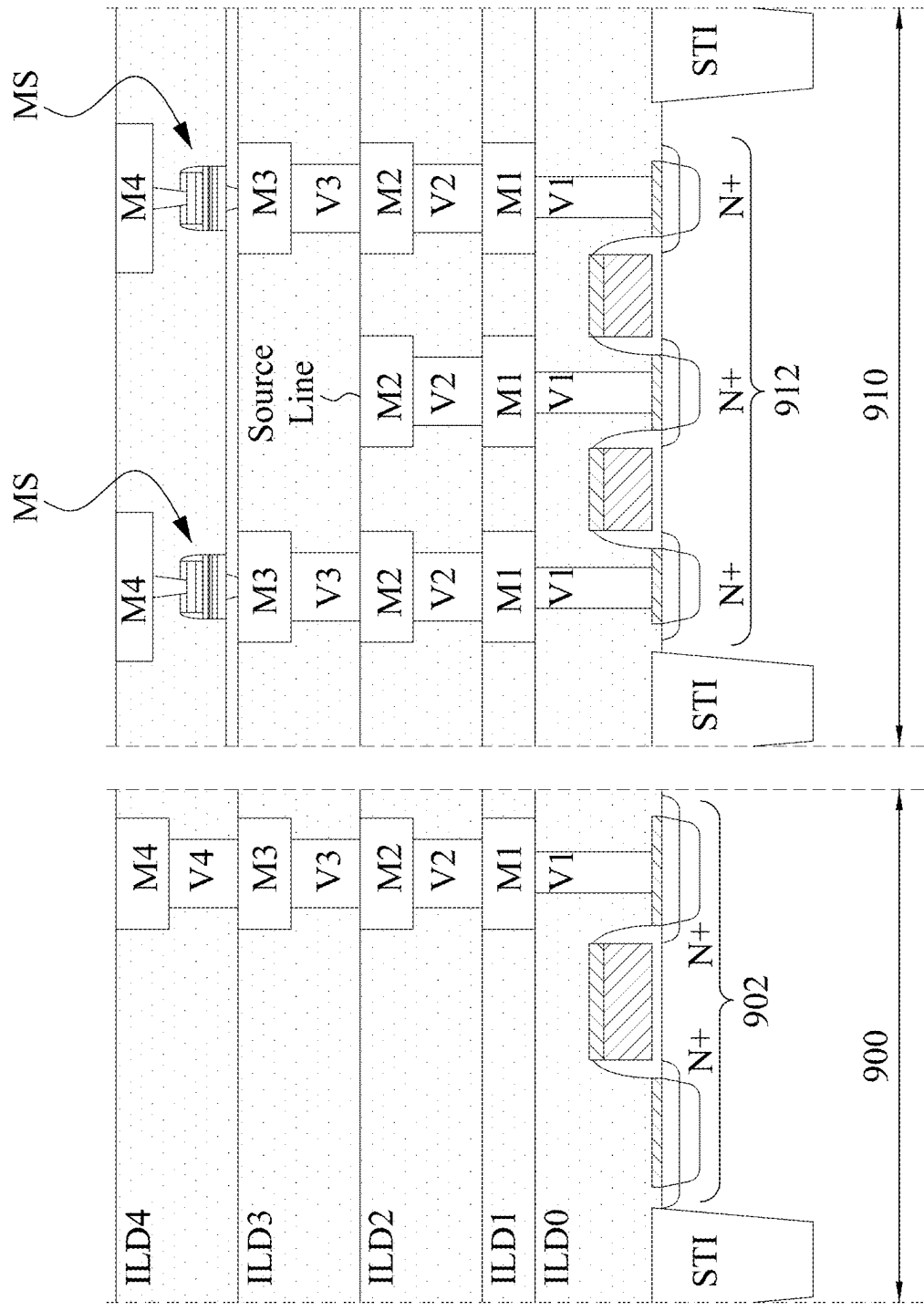


Fig. 18

MEMORY DEVICE WITH BOTTOM ELECTRODE

PRIORITY CLAIM AND CROSS-REFERENCE

[0001] The present application is a continuation application of U.S. application Ser. No. 18/333,145, filed Jun. 12, 2023, which is a divisional application of U.S. application Ser. No. 16/912,341, filed Jun. 25, 2020, now U.S. Pat. No. 11,723,294, issued Aug. 8, 2023, all of which are herein incorporated by reference in their entireties.

BACKGROUND

[0002] In integrated circuit (IC) devices, resistive random access memory (RRAM) is an emerging technology for next generation non-volatile memory devices. RRAM is a memory structure including an array of RRAM cells each of which stores a bit of data using resistance values, rather than electronic charge. Particularly, RRAM cell includes a resistance switch layer, the resistance of which can be adjusted to represent logic “0” or logic “1.”

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.

[0004] FIGS. 1-8 illustrate various stages in the formation of a memory device according to some embodiments of the present disclosure.

[0005] FIG. 9 is a cross-sectional view of a memory device in accordance with some embodiments of the present disclosure.

[0006] FIG. 10 is a cross-sectional view of a memory device in accordance with some embodiments of the present disclosure.

[0007] FIGS. 11-17 illustrate various stages in the formation of a memory device according to some embodiments of the present disclosure.

[0008] FIG. 18 is a cross-sectional view of a memory device in accordance with some embodiments of the present disclosure.

DETAILED DESCRIPTION

[0009] The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.

[0010] Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature’s relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.

[0011] Resistive random-access memory (RRAM) devices have a bottom electrode that is separated from an overlying top electrode by a dielectric data storage layer having a variable resistance. RRAM devices are configured to store data based on a resistive state of the dielectric data storage layer. For example, the dielectric data storage layer may have a high resistance state associated with a first data state (e.g., a ‘0’) or a low resistance state associated with a second data state (e.g., a ‘1’).

[0012] During operation of an RRAM device, bias voltages are applied to the bottom and top electrodes to reversibly change a resistive state of the dielectric data storage layer. The bias voltages change the resistive state of the dielectric data storage layer by controlling the movement of oxygen between the electrodes and the dielectric data storage layer to either form or break conductive filaments extending through the dielectric data storage layer. For example, a first set of bias voltages may induce conductive paths/filaments (e.g., chains of oxygen vacancies) to form across the dielectric data storage layer to achieve a low resistance state, while a second set of bias voltages may break conductive paths/filaments within the dielectric data storage layer to achieve a high resistance state.

[0013] It has been appreciated that when forming a conductive filament to achieve a low resistive state, the bias voltages may cause oxygen from the dielectric data storage layer to move to deep within a top electrode and/or within an overlying layer. However, if oxygen moves far away from the dielectric data storage layer, it can be difficult to pull the oxygen back to dielectric data storage layer to subsequently break the conductive filament. As an RRAM device is operated over many cycles, the amount of oxygen moved to deep within the top electrode and/or the overlying layer increases, which in turn will damage the dielectric data storage layer and result in hard reset bit (HRB) issue, eventually leading to RRAM failure.

[0014] Embodiments of the present disclosure relates to a RRAM device having a dielectric data storage layer formed over a treated surface, in which the treated surface has an increased amount of oxide vacancies. By increasing the amount of oxide vacancies of the surface, the amount of oxide vacancies in the dielectric data storage layer is increased, such that the oxygen can be kept close to the dielectric data storage layer and a reliability of the RRAM device can be improved.

[0015] A RRAM device and the method of forming the same are provided in accordance with various exemplary embodiments. The intermediate stages of forming the RRAM device are illustrated. The variations of the embodiments are discussed. Throughout the various views and illustrative embodiments, like reference numbers are used to designate like elements.

[0016] FIGS. 1-8 illustrate various stages in the formation of a memory device according to some embodiments of the present disclosure. The illustration is merely exemplary and is not intended to limit beyond what is specifically recited in the claims that follow. It is understood that additional operations may be provided before, during, and after the operations shown by FIGS. 1-8, and some of the operations described below can be replaced or eliminated for additional embodiments of the method. The order of the operations/processes may be interchangeable.

[0017] FIG. 1 illustrates a semiconductor substrate having transistors and one or more metal/dielectric layers **110** thereon. The semiconductor substrate may be a silicon substrate. Alternatively, the substrate may include another elementary semiconductor, such as germanium; a compound semiconductor including silicon carbide; an alloy semiconductor including silicon germanium; or combinations thereof. In some embodiments, the substrate is a semiconductor on insulator (SOI) substrate. The substrate may include doped regions, such as p-wells and n-wells. The transistors are formed by suitable transistor fabrication processes and may be a planar transistor, such as polysilicon gate transistors or high-k metal gate transistors, or a multi-gate transistor, such as fin field effect transistors. After the transistors are formed, one or more metal/dielectric layers **110** of a multi-level interconnect (MLI) is formed over the transistors.

[0018] The metal/dielectric layer **110** includes one or more conductive features **112** embedded in an inter-layer dielectric (ILD) layer **114**. The ILD layer **114** may be silicon oxide, fluorinated silica glass (FSG), carbon doped silicon oxide, tetra-ethyl-ortho-silicate (TEOS) oxide, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), Black Diamond® (Applied Materials of Santa Clara, Calif.), amorphous fluorinated carbon, low-k dielectric material, the like or combinations thereof. The conductive features **112** may be aluminum, aluminum alloy, copper, copper alloy, titanium, titanium nitride, tantalum, tantalum nitride, tungsten, cobalt, the like, and/or combinations thereof. The substrate may also include active and passive devices, for example, underlying the metal/dielectric layers **110**. These further components are omitted from the figures for clarity.

[0019] A dielectric layer **120** may be formed on the metal/dielectric layer **110**. The dielectric layer **120** may be silicon carbide, silicon oxynitride, silicon nitride, carbon doped silicon nitride or carbon doped silicon oxide. The dielectric layer **120** may include one or plural layers. In some embodiments, a material of a top layer of the dielectric layer **120** is selected to have a chemical mechanical polishing (CMP) resistance property different from that of a bottom electrode material formed in subsequent processes. The dielectric layer **120** is deposited over the metal/dielectric layer **110** using a chemical vapor deposition (CVD) process such as plasma enhanced (PE) CVD, high-density plasma (HDP) CVD, inductively-coupled-plasma (ICP) CVD, or thermal CVD.

[0020] Bottom electrode vias (BEVA) **130** are formed in the dielectric layer **120**. An exemplary formation method of the BEVAs **130** includes etching an opening **O1** in the dielectric layer **120** and exposing a portion of the conductive feature **112**, and filling the opening **O1** with suitable conductive materials, thereby forming the BEVA **130**. In some embodiments, after filling the opening **O1** with the materials, a planarization process, such as a chemical-mechanical

polish (CMP) process, is performed to remove excess conductive materials outside the opening **O1**. In some embodiments, the BEVA is electrically connected to an underlying electrical component, such as a transistor, through the conductive feature **112**.

[0021] In some embodiments, the BEVA **130** is a multi-layered structure and includes, for example, a diffusion barrier layer and a filling metal filling a recess in the diffusion barrier layer. In some embodiments, the diffusion barrier layer is a titanium nitride (TiN) layer or a tantalum nitride (TaN) layer, which can act as a suitable barrier to prevent metal diffusion. Formation of the diffusion barrier layer may be exemplarily performed using CVD, PVD, ALD, the like, and/or a combination thereof. In some embodiments, the filling metal is titanium (Ti), tantalum (Ta), platinum (Pt), ruthenium (Ru), tungsten (W), aluminum (Al), copper (Cu), TiN, TaN, the like, and/or combinations thereof. Formation of the filling metal may be exemplarily performed using CVD, PVD, ALD, the like, and/or a combination thereof.

[0022] A bottom electrode stack layer **140** is then formed over the BEVA **130** and over the dielectric layer **120**, so that the bottom electrode stack layer **140** extends along a top surface of the BEVA **130** and top surfaces of the dielectric layer **120**. The bottom electrode stack layer **140** can be a single-layered structure or a multi-layered structure. For example, the bottom electrode stack layer **140** includes a first electrode layer **142**, a second electrode layer **144** over the first electrode layer **142**, and a buffer layer **146** over the second electrode layer **144**.

[0023] In some embodiments, the first electrode layer **142** may include titanium (Ti), tantalum (Ta), platinum (Pt), ruthenium (Ru), tungsten (W), aluminum (Al), copper (Cu), TiN, TaN, the like, and/or a combination thereof. Formation of the first electrode layer **142** may be exemplarily performed using CVD, PVD, ALD, the like, and/or a combination thereof.

[0024] In some embodiments, the second electrode layer **144** is formed over the first electrode layer **142**. The second electrode layer **144** may include a material different from that of the first electrode layer **142**. For example, the second electrode layer **144** may include Ru, Ti, W, Ni, Al, Pd, Co, or the combination thereof. In some embodiments, the second electrode layer **144** may be more inactive than the first electrode layer **142**. For example, the second electrode layer **144** may be more inert to oxygen than the first electrode layer **142** is. In some embodiments, the second electrode layer **144** may include noble metals, while the first electrode layer **142** may include non-noble metals. For example, the second electrode layer **144** may include Ru, Pd, or the like, and the first electrode layer **142** may include Ti, Ta, Al, W, TiN, TaN or the like. The second electrode layer **144** may be deposited by ALD. Alternatively, the second electrode layer **144** is deposited by an electroless plating process or other suitable process.

[0025] In some embodiments, the buffer layer **146** is formed over the second electrode layer **144**. The buffer layer **146** may include a material different from that of the second electrode layer **144**. In some embodiments, the buffer layer **146** may be more active than the second electrode layer **144**. For example, in some embodiments, the buffer layer **146** may include non-noble metals, while the second electrode layer **144** may include noble metals. For example, the buffer layer **146** may include tantalum, TaN, or the combination

thereof. The buffer layer **146** may be deposited by ALD. Alternatively, the buffer layer **146** is deposited by an electroless plating process or other suitable process. The buffer layer **146** may have a thickness in a range of about 5 angstroms to about 20 angstroms. If the thickness of the buffer layer **146** is greater than about 20 angstroms, a forming voltage for triggering the RRAM device may be too large to be bear by the gate oxide of a logic device, which derives the demands of high voltage logic devices, which may occupy more chip area. If the thickness of the buffer layer **146** is less than about 5 angstroms, the subsequent surface treatment performed to the top surface **146T** of the buffer layer **146** may not make the top surface **146T** include defects, which in turn will fail the improvement on the cycling of the resistance switch layer **150** (referring to FIG. 3) to be formed.

[0026] Reference is made to FIG. 2. A surface treatment is performed to a top surface **146T** of the buffer layer **146** by inducing a gas or plasma to the top surface **146T** of the buffer layer **146**. The gas or plasma may include oxide-containing gas or nitrogen-containing gas. For example, the gas or plasma may include O_2 or N_2O . In some embodiments, the surface treatment may be performed in ex-situ chamber after the formation of the buffer layer **146**. After the surface treatment, the top surface **146T** of the buffer layer **146** may include defects, such as dangling bonds or oxide vacancies. In other word, the surface treatment may create oxide vacancies over a top surface of the bottom electrode stack layer **140**.

[0027] In some embodiments, the surface treatment may turn a top portion of the buffer layer **146** adjoining the top surface **146T** into a metal-containing compound layer. For example, the surface treatment may oxidize a top portion of the buffer layer **146** adjoining the top surface **146T**. The oxidized top portion of the buffer layer **146** may be referred to as a metal-containing oxide layer **146P** hereinafter. The metal-containing oxide layer **146P** may have a same metal as that included in the buffer layer **146**. The metal included in the metal-containing oxide layer **146P** and the buffer layer **146** may be a non-noble metal. For example, while the buffer layer **146** include tantalum or TaN, the metal-containing oxide layer **146P** may include TaO. A thickness of the metal-containing oxide layer **146P** may be several angstroms, for example, in a range from about 3 angstroms to about 8 angstroms. In some embodiments, the metal-containing oxide layer **146P** is observable by TEM and/or EDX analysis. Sometimes, in alternative embodiments, the metal-containing oxide layer **146P** may be too thinned to be observed.

[0028] In some embodiments, the surface treatment using the oxide-containing gas (e.g., O_2) is performed in a chemical vapor deposition (CVD) chamber with a power ranging from about 5 W to about 800 W and a time duration ranging from about 3 seconds to about 50 seconds. If the power is greater than about 800 W or the time duration is greater than about 50 seconds, the resistance switch layer **150** (referring to FIG. 3) to be formed may have leakage issues, which may fail the RRAM device. If the power is less than about 5 W or the time duration is less than about 3 seconds, the cycling of the resistance switch layer **150** (referring to FIG. 3) to be formed may not be effectively improved.

[0029] In some embodiments, the surface treatment using the nitrogen-containing gas (e.g., N_2O) is performed in a chemical vapor deposition (CVD) chamber with a power

ranging from about 5 W to about 600 W and a time duration ranging from about 3 seconds to about 50 seconds. If the power is greater than about 600 W or the time duration is greater than about 50 seconds, the resistance switch layer **150** (referring to FIG. 3) to be formed may have leakage issues, which may fail the RRAM device. If the power is less than about 5 W or the time duration is less than about 3 seconds, the cycling of the resistance switch layer **150** (referring to FIG. 3) to be formed may not be effectively improved.

[0030] Reference is made to FIG. 3. A resistance switch layer **150**, a capping layer **160**, a top electrode layer **170**, and a hardmask layer **180** are subsequently formed on the top surface **146T**. In some embodiments where the metal-containing oxide layer **146P** is observable, the resistance switch layer **150** may be in contact with the metal-containing oxide layer **146P**. The deposited resistance switch layer **150** may be spaced apart from the buffer layer **146** of the bottom electrode stack layer **140** by the metal-containing oxide layer **146P**. In some embodiments where the metal-containing oxide layer **146P** is unobservable, the resistance switch layer **150** may be in contact with the buffer layer **146**. The resistance switch layer **150** includes a material having a variable resistance configured to undergo a reversible phase change between a high resistance state and a low resistance state. For example, the resistance switch layer **150** may include high-k dielectric films. In some embodiments, the resistance switch layer **150** is a metal oxide, which may be hafnium oxide, zirconium oxide, aluminum oxide, nickel oxide, tantalum oxide, titanium oxide, and other oxides used as a resistance switch layer. The metal oxide may have a non-stoichiometric oxygen to metal ratio. Depending on the method of deposition, the oxygen to metal ratio and other process conditions may be tuned to achieve specific resistance switch layer **150** properties. For example, a set of conditions may yield a low 'forming' voltage and another set of conditions may yield a low 'read' voltage. The metal oxide may be deposited. In some embodiments, the metal oxide is a transition metal oxide. In other embodiments, the resistance switch layer is a metal oxynitride.

[0031] The resistance switch layer **150** may be formed by a suitable technique, such as atomic layer deposition (ALD) with a precursor containing a metal and oxygen. Other chemical vapor deposition (CVD) techniques may be used. In another example, the resistance switch layer **150** may be formed by a physical vapor deposition (PVD), such as a sputtering process with a metallic target and with a gas supply of oxygen and optionally nitrogen to the PVD chamber. In yet another example, the resistance switch layer **150** may be formed by an electron-beam deposition process.

[0032] In some cases, when a formed RRAM device is operated over many cycles, oxygen may move far away from the resistance switch layer **150**, and it can be difficult to pull the oxygen back to the resistance switch layer **150** to subsequently break the conductive filament, which in turn will damage the dielectric data storage layer and result in hard reset bit (HRB) issue, eventually leading to RRAM failure.

[0033] In some embodiments of the present disclosure, by increasing the amount of oxide vacancies of the surface **146T** where the resistance switch layer **150** is formed on, the amount of oxide vacancies in the resistance switch layer **150** is increased. Through the configuration, when a formed RRAM device is operated, it can be easier to pull the oxygen

back to the resistance switch layer **150** to subsequently break the conductive filament, and therefore improving the reliability of the RRAM device.

[0034] In various embodiments, the capping layer **160** over the resistance switch layer **150** is a metal, for example, titanium, hafnium, platinum, ruthenium or tantalum. In some embodiments, the capping layer may include hafnium oxide, aluminum oxide, tantalum oxides, other metal oxidation composite layers, or the combination thereof. The capping layer **160** may be deposited using a PVD process, a CVD, or an ALD process.

[0035] The top electrode layer **170** may be metal, metal-nitride, doped polysilicon or other suitable conductive material. For example, the top electrode layer **170** may be tantalum nitride, titanium nitride, titanium, tantalum or platinum. The top electrode layer **170** may be single or bilayer. The top electrode layer **170** may be formed by PVD, CVD, ALD, or other suitable technique. Alternatively, the top electrode layer **170** includes other suitable conductive material to electrically connect the device to other portion of an interconnect structure. In some embodiments, the capping layer **160** and the top electrode layer **170** may be formed of the same material, but using different processes so as to vary a specific material property. In other embodiments, the capping layer **160** is a metal and the top electrode layer **170** is a metal nitride, for example, the capping layer **160** may be titanium and the top electrode layer **170** may be a tantalum nitride.

[0036] The hardmask layer **180** may be made of silicon nitride, silicon carbide, or other composite dielectric layers. In some embodiments, a silicon oxynitride is used. Silicon oxynitride has a good etch selectivity against the bottom electrode metal. Other hardmask material including silicon carbide, carbon-doped silicon nitride, or silicon nitride may be used.

[0037] Reference is made to FIG. 4. The hardmask layer **180**, the top electrode layer **170**, and the capping layer **160** (referring to FIG. 3) are patterned into a hardmask **182**, a top electrode **172**, a capping layer **162**, respectively. The patterning process may include a photolithography operation where a photoresist is deposited over the hardmask layer **180** (referring to FIG. 3), a pattern is defined by exposing photoresist to a radiation, and developing the photoresist to create a photoresist pattern. The photoresist pattern is then used as an etch mask to protect desired portions of the hardmask layer **180** (referring to FIG. 3). The hardmask layer **180** (referring to FIG. 3) may then be patterned using an etching operation. In some embodiments, an etchant used to pattern the hardmask layer **180** (referring to FIG. 3) includes an etching chemistry including gases of CF_4 , CH_2F_2 and/or other chemicals. The photoresist mask is removed after the patterning. In some embodiments, the photoresist mask can be removed by adding oxygen to the etchant. Subsequently, the hardmask **182** is used as an etchmask to pattern the top electrode layer **170** and the capping layer **160**. In some embodiments, an etchant is applied to etch an exposed portion of the top electrode layer **170** and the capping layer **160** that is not covered by the hardmask **182**. The etch process stops when the resistance switch layer **150** is reached. Techniques are available to detect the end of etching when a new material layer is reached so as to reduce the amount of over etching.

[0038] Reference is made to FIG. 5. A spacer layer **190** is deposited over the hardmask **182** and the resistance switch

layer **150**. The spacer layer **190** may be made of silicon nitride, silicon oxynitride, and silicon oxide. The spacer layer **190** may be formed by conformally coating a spacer material covering the top and sidewalls of the top electrode **172** and the capping layer **162**.

[0039] Reference is made to FIG. 6. An anisotropic etch process is performed to remove horizontal portions of the spacer layer **190** (referring to FIG. 5), and remain vertical portions of the spacer layer **190**, thereby forming the spacer **192**. The spacer **192** surrounds the hardmask **182**, the top electrode **172**, and the capping layer **162**, and thus protects them against subsequent etch operations. The height and width of the spacer **192** after etching may be tuned by adjusting deposition and etching parameters.

[0040] According to various embodiments, the spacer etching is performed without patterning using a patterned mask because the shape of the conformal spacer material can be etched to form the spacer **192**. However, other spacer shapes may be formed by patterning the spacer material using a patterned mask. If a patterned mask is used, the spacer **192** may be formed of suitable shapes. For example, the spacer **192** may include a portion over the hardmask **182** and the top corners of the hardmask **182** so as to further protect the memory structure during bottom electrode etch.

[0041] Then, reference is made to FIG. 7. The resistance switch layer **150**, the metal-containing oxide layer **146P**, the buffer layer **146**, the second electrode layer **144**, and the first electrode layer **142** (referring to FIG. 6) are patterned into a resistance switch element **152**, a metal-containing oxide portion **146P'**, a buffer element **146'**, a second electrode **144'**, and a first electrode **142'**, respectively. The spacer **192** and the hardmask **182** are used as an etch mask to remove portions of the resistance switch layer **150**, the buffer layer **146**, the second electrode layer **144**, and the first electrode layer **142** (referring to FIG. 6). In some embodiments, the buffer element **146'**, the second electrode **144'**, and the first electrode **142'** in combination may be referred to as a bottom electrode **140'**. Through the operations, a memory structure MS is formed, and the memory structure MS includes the bottom electrode **140'**, the metal-containing oxide portion **146P'** over the bottom electrode **140'**, the resistance switch element **152** over the metal-containing oxide portion **146P'**, the capping layer **162** over the resistance switch element **152**, the top electrode **172** over the capping layer **162**, and the hardmask **182** over the top electrode **172**.

[0042] In the present embodiments, the bottom electrode stack layer **140** (referring to FIG. 6) is patterned into the bottom electrode **140'** after patterning the resistance switch layer **150** and the top electrode layer **170** (referring to FIG. 5) into the resistance switch element **152** and the top electrode **172**. In some alternative embodiments, the bottom electrode stack layer **140** (referring to FIG. 5) may be patterned into the bottom electrode **140'** prior to patterning the resistance switch layer **150** and the top electrode layer **170** (referring to FIG. 5) into the resistance switch element **152** and the top electrode **172**. For example, the bottom electrode stack layer **140** (referring to FIG. 5) may be patterned into the bottom electrode **140'**, and then the resistance switch layer **150** and the top electrode layer **170** (referring to FIG. 5) are deposited over the bottom electrode **140'** in some embodiments.

[0043] Reference is made to FIG. 8. An inter-layer dielectric layer **200** is deposited over the memory structure MS and the metal/dielectric layer **110** using suitable deposition

techniques. The inter-layer dielectric layer **200** may be silicon oxide, extreme or extra low-k silicon oxide such as a porous silicon oxide layer, or other commonly used inter-layer dielectric material. After the formation of the inter-layer dielectric layer **200**, a top electrode via **2000** are etched in the inter-layer dielectric layer **200** to expose a top electrode **172**. Subsequently, the top electrode via **2000** is filled with a conductive feature **210**, such as a metal. The filling may also include one or more liner and barrier layers in addition to a metal conductor. The liner and/or barrier may be conductive and deposited using CVD or PVD. The metal may be deposited using PVD or one of the plating methods, such as electrochemical plating. After the filling, a planarization process, such as chemical mechanical polishing (CMP), is performed to remove excess conductive feature **210**.

[0044] A metal/dielectric layer **230** may be formed over the memory structure MS and the inter-layer dielectric layer **200**, and the metal/dielectric layer **230** may include one or more conductive features **232** embedded in an inter-layer dielectric (ILD) layer **234**. The ILD layer **234** may be silicon oxide, fluorinated silica glass (FSG), carbon doped silicon oxide, tetra-ethyl-ortho-silicate (TEOS) oxide, phosphosilicate glass (PSG), borophosphosilicate glass (BPSG), Black Diamond® (Applied Materials of Santa Clara, Calif.), amorphous fluorinated carbon, low-k dielectric material, the like or combinations thereof. The conductive features **232** may be aluminum, aluminum alloy, copper, copper alloy, titanium, titanium nitride, tantalum, tantalum nitride, tungsten, cobalt, the like, and/or combinations thereof.

[0045] In some embodiments of the present disclosure, the surface treatment to the surface **146T** may increase the amount of oxide vacancies in the resistance switch element **152** thereby improving a reliability of the memory structure MS. In some embodiments, the surface treatment to the surface **146T** may further enlarge the difference between currents measured at the first data state (e.g., a '0') and the second data state (e.g., a '1'), which in turn may improve the operation window of the memory device.

[0046] FIG. 9 is a cross-sectional view of a memory device in accordance with some embodiments of the present disclosure. The details of the present embodiments are similar to that of the embodiments of FIGS. 1-8, except the configuration of the bottom electrode **140'**. In the present embodiments, the bottom electrode **140'** may not include the second electrode **144'**. For example, in the present embodiments, a bottom surface of the buffer element **146'** is directly in contact with a top surface of the first electrode **142'**. Other details of the present embodiments are similar to those described above, and therefore not repeated herein.

[0047] FIG. 10 is a cross-sectional view of a memory device in accordance with some embodiments of the present disclosure. The configuration of the present embodiments are similar to that of the embodiments of FIGS. 1-8, except that the surface treatment performed to the top surface **146T** of the buffer layer **146** (referring to FIG. 1) may not form an observable metal-containing oxide portion. That is, the formed memory structure MS may not include an observable metal-containing oxide portion that has a same metal as that included in the buffer element **146'** between the buffer element **146'** of the bottom electrode **140'** and the resistance switch element **152**. For example, in the present embodiments, a bottom surface of the resistance switch element **152** is directly in contact with a top surface of the buffer element

146', without a TaO layer therebetween. Other details of the present embodiments are similar to those described above, and therefore not repeated herein.

[0048] FIGS. 11-17 illustrate various stages in the formation of a memory device according to some embodiments of the present disclosure. The configurations of the present embodiments are similar to that of the embodiments of FIGS. 1-8, except the shapes of the stacked layers. For example, the stacked layers have a recess profile corresponding to the openings **O1** in the dielectric layer **120**. The illustration is merely exemplary and is not intended to limit beyond what is specifically recited in the claims that follow. It is understood that additional operations may be provided before, during, and after the operations shown by FIGS. 11-17, and some of the operations described below can be replaced or eliminated for additional embodiments of the method. The order of the operations/processes may be interchangeable.

[0049] FIG. 11 illustrates a semiconductor substrate **102** having transistors and one or more metal/dielectric layers **110** thereon. The semiconductor substrate **102** may be a silicon substrate. Alternatively, the substrate **102** may include another elementary semiconductor, such as germanium; a compound semiconductor including silicon carbide; an alloy semiconductor including silicon germanium; or combinations thereof. In some embodiments, the substrate **102** is a semiconductor on insulator (SOI) substrate. The substrate **102** may include doped regions, such as p-wells and n-wells. The transistors are formed by suitable transistor fabrication processes and may be a planar transistor, such as polysilicon gate transistors or high-k metal gate transistors, or a multi-gate transistor, such as fin field effect transistors. After the transistors are formed, one or more metal/dielectric layers **110** of a multi-level interconnect (MLI) is formed over the transistors. The metal/dielectric layer **110** includes one or more conductive features **112** embedded in an inter-layer dielectric (ILD) layer **114**.

[0050] In the present embodiments, a dielectric layer **120** is formed on the metal/dielectric layer **110**, and an opening **O1** is etched in the dielectric layer **120** to expose a portion of the conductive feature **112** in the metal/dielectric layer **110**. The dielectric layer **120** may include suitable dielectric material, such as silicon oxide.

[0051] Subsequently, a bottom electrode stack layer **140** is deposited over the dielectric layer **120** and filling the opening **O1**. In some embodiments, the bottom electrode stack layer **140** can be a single-layered structure or a multi-layered structure. For example, the bottom electrode stack layer **140** includes a diffusion barrier layer **141**, a first electrode layer **142** over the diffusion barrier layer **141**, a second electrode layer **144** over the first electrode layer **142**, and a buffer layer **146** over the second electrode layer **144**.

[0052] In some embodiments, the diffusion barrier layer **141** is a titanium nitride (TiN) layer or a tantalum nitride (TaN) layer, which can act as a suitable barrier to prevent metal diffusion. Formation of the diffusion barrier layer **141** may be exemplarily performed using CVD, PVD, ALD, the like, and/or a combination thereof.

[0053] In some embodiments, the first electrode layer **142** may include titanium (Ti), tantalum (Ta), platinum (Pt), ruthenium (Ru), tungsten (W), aluminum (Al), copper (Cu), TiN, TaN, the like, and/or combinations thereof. Formation

of the first electrode layer 142 may be exemplarily performed using CVD, PVD, ALD, the like, and/or a combination thereof.

[0054] In some embodiments, the second electrode layer 144 may include Ru, Ti, W, Ni, Al, Pd, or Co, or the combination thereof. In some embodiments, the second electrode layer 144 may be more inactive than the first electrode layer 142. For example, the second electrode layer 144 may be more inert to oxygen than the first electrode layer 142 is. In some embodiments, the second electrode layer 144 may include noble metals, while the first electrode layer 142 may include non-noble metals. For example, the second electrode layer 144 may include Ru, Pd, or the like, and the first electrode layer 142 may include Ti, Ta, Al, W, TiN, TaN or the like. The second electrode layer 144 may be deposited by ALD. Alternatively, the second electrode layer 144 is deposited by an electroless plating process or other suitable process.

[0055] In some embodiments, the buffer layer 146 is formed over the second electrode layer 144. In some embodiments, the buffer layer 146 may be more active than the second electrode layer 144. For example, in some embodiments, the buffer layer 146 may include non-noble metals, while the second electrode layer 144 may include noble metals. For example, the buffer layer 146 may include tantalum, TaN, or the combination thereof. The buffer layer 146 may be deposited by ALD. Alternatively, the buffer layer 146 is deposited by an electroless plating process or other suitable process.

[0056] In the present embodiments, the diffusion barrier layer 141, the first and second electrode layer 142 and 144, and the buffer layer 146 of the bottom electrode stack layer 140 have profiles conforming to the opening O1 in the dielectric layer 120. For example, each of the diffusion barrier layer 141, the first and second electrode layer 142 and 144, and the buffer layer 146 has a first portion in the opening O1 in the dielectric layer 120 and a second portion over a top surface of the dielectric layer 120. In other word, each of the diffusion barrier layer 141, the first and second electrode layer 142 and 144, and the buffer layer 146 has a recess above the opening O1 in the dielectric layer 120.

[0057] Reference is made to FIG. 12. A surface treatment is performed to a top surface 146T of the buffer layer 146 by inducing a gas or plasma to the top surface 146T of the buffer layer 146. The gas or plasma may include oxide-containing gas or nitrogen-containing gas. For example, the gas or plasma may include O₂ or N₂O. In some embodiments, the surface treatment may be performed in ex-situ chamber after the formation of the buffer layer 146. After the surface treatment, the top surface 146T of the buffer layer 146 may include defects, such as dangling bonds or oxide vacancies. In other word, the surface treatment may create oxide vacancies over a top surface of the bottom electrode stack layer 140.

[0058] In some embodiments, the surface treatment may oxidize a top portion of the buffer layer 146 adjoining the top surface 146T. The oxidized top portion of the buffer layer 146 may be referred to as a metal-containing oxide layer 146P hereinafter. The metal-containing oxide layer 146P may have a same metal as that included in the buffer layer 146. For example, while the buffer layer 146 include tantalum or TaN, the metal-containing oxide layer 146P may include TaO. In the present embodiments, the metal-containing oxide layer 146P may have a profile conforming to the

opening O1 in the dielectric layer 120. For example, the metal-containing oxide layer 146P has a first portion in the opening O1 in the dielectric layer 120 and a second portion over the top surface of the dielectric layer 120. In other word, the metal-containing oxide layer 146P has a recess above the opening O1 in the dielectric layer 120. In some embodiments, a bottom surface of the metal-containing oxide layer 146P is higher than a top surface of the dielectric layer 120.

[0059] Reference is made to FIG. 13. A resistance switch layer 150, a capping layer 160, a top electrode layer 170, and a hardmask layer 180 are subsequently formed on the top surface 146T. In some embodiments, the resistance switch layer 150 is a metal oxide, which may be hafnium oxide, zirconium oxide, aluminum oxide, nickel oxide, tantalum oxide, titanium oxide, and other oxides used as a resistance switch layer. The metal oxide may have a non-stoichiometric oxygen to metal ratio. Depending on the method of deposition, the oxygen to metal ratio and other process conditions may be tuned to achieve specific resistance switch layer 150 properties. For example, a set of conditions may yield a low 'forming' voltage and another set of conditions may yield a low 'read' voltage. In some embodiments of the present disclosure, by increasing the amount of oxide vacancies of the surface 146T where the resistance switch layer 150 is formed on, the amount of oxide vacancies in the resistance switch layer 150 is increased, thereby improving a reliability of the RRAM device to be formed. Other details regarding the formation of these layers are similar to those illustrated above, and therefore not repeated herein.

[0060] Reference is made to FIG. 14. The hardmask layer 180, the top electrode layer 170, and the capping layer 160 (referring to FIG. 13) are patterned into a hardmask 182, a top electrode 172, a capping layer 162, respectively. The patterning process may include suitable photolithography and etching operations. Other details regarding the patterning process are similar to those illustrated above, and therefore not repeated herein.

[0061] Reference is made to FIG. 15. A spacer 192 is formed to surround the hardmask 182, the top electrode 172, and the capping layer 162, and thus protects them against subsequent etch operations. The spacer 192 may be made of silicon nitride, silicon oxynitride, and silicon oxide. Formation of the spacer 192 may include depositing a spacer layer over the structure of FIG. 14, and then removing portions of the spacer layer by anisotropic etching process. Other details regarding the formation of the spacer 192 are illustrated above, and therefore not repeated herein.

[0062] Reference is made to FIG. 16. The resistance switch layer 150, the metal-containing oxide layer 146P, the buffer layer 146, the second electrode layer 144, the first electrode layer 142, and the diffusion barrier layer 141 (referring to FIG. 15) are patterned into a resistance switch element 152, a metal-containing oxide portion 146P', a buffer element 146', a second electrode 144', a first electrode 142', and a diffusion barrier layer 141', respectively. The spacer 192 and the hardmask 182 are used as an etch mask to remove portions of the resistance switch layer 150, the buffer layer 146, the second electrode layer 144, the first electrode layer 142, and the diffusion barrier layer 141 (referring to FIG. 15). In some embodiments, the buffer element 146', the second electrode 144', the first electrode 142', and the diffusion barrier layer 141' in combination may be referred to as a bottom electrode 140'. Through the

operations, a memory structure MS is formed, and the memory structure MS includes the bottom electrode **140'**, the resistance switch element **152**, the capping layer **162**, the top electrode **172**, and the hardmask **182**.

[0063] In the present embodiments, the bottom electrode **140'** has a via portion **140VP** in the dielectric layer **120** and a top portion **140TP** over a top surface of the dielectric layer **120**. The metal-containing oxide portion **146P'** has a first portion **P1** over the via portion **140VP** of the bottom electrode **140'** and a second portion **P2** over the top portion **140TP** of the bottom electrode **140'**, and a top surface of the second portion **P2** of the metal-containing oxide portion **146P'** is higher than a top surface of the first portion **P1** of the metal-containing oxide portion **146P'**. In some embodiments, a bottom surface of the first portion **P1** of the metal-containing oxide portion **146P'** is higher than a top surface of the dielectric layer **120**.

[0064] Reference is made to FIG. 17. An inter-layer dielectric layer **200** is deposited over the memory structure MS and the metal/dielectric layer **110** using suitable deposition techniques. After the formation of the inter-layer dielectric layer **200**, a top electrode via **2000** is etched in the inter-layer dielectric layer **200** to expose a top electrode **172**. Subsequently, the top electrode via **2000** is filled with a conductive feature **210**, such as a metal. The filling may also include one or more liner and barrier layers in addition to a metal conductor. A metal/dielectric layer **230** may be formed over the memory structure MS, and the metal/dielectric layer **230** may include one or more conductive features **232** embedded in an inter-layer dielectric (ILD) layer **234**. Other details of the present embodiments are similar to those illustrated above, and therefore not repeated herein.

[0065] FIG. 18 is a cross-sectional view of a semiconductor device in accordance with some embodiments of the present disclosure. The semiconductor device includes a logic region **900** and a memory region **910**. Logic region **900** may include circuitry, such as an exemplary logic transistor **902**, for processing information received from the memory structures MS in the memory region **910** and for controlling reading and writing functions of the memory structures MS. In some embodiments, the memory structures MS may be similar to those shown above.

[0066] As depicted, the semiconductor device is fabricated using four metallization layers, labeled as M1 through M4, with four layers of metallization vias or interconnects, labeled as V1 through V4. Other embodiments may contain more or fewer metallization layers and a corresponding more or fewer number of vias. Logic region **900** includes a full metallization stack, including a portion of each of metallization layers M1-M4 connected by interconnects V2-V4, with the interconnect V1 connecting the stack to a source/drain contact of logic transistor **902**. The memory region **910** includes a full metallization stack connecting memory structures MS to transistors **912** in the memory region **910**, and a partial metallization stack connecting a source line to transistors **912** in the memory region **910**. Memory structures MS are depicted as being fabricated in between the top of the metallization layer M3 and the bottom of the metallization layer M4. Also included in semiconductor device is a plurality of ILD layers. Five ILD layers, identified as ILD0 through ILD4 are depicted in FIG. 18 as spanning the logic region **900** and the memory region **910**. The ILD layers may provide electrical insulation as well as

structural support for the various features of the semiconductor device during many fabrication process steps.

[0067] Based on the above discussions, it can be seen that the present disclosure offers advantages. It is understood, however, that other embodiments may offer additional advantages, and not all advantages are necessarily disclosed herein, and that no particular advantage is required for all embodiments. One advantage is that the amount of oxide vacancies in the resistance switch layer is increased by the surface treatment to its underlying layer (e.g., the bottom electrode), such that it can be easier to pull the oxygen back to the resistance switch layer to subsequently break the conductive filament, thereby improving the cycle reliability of the RRAM device. Another advantage is that the surface treatment further enlarges the difference between currents measured at the first data state (e.g., a '0') and the second data state (e.g., a '1'), which in turn may improve the operation window of the memory device. In some embodiments, the surface treatment to the underlying layer may form an observable metal-containing oxide portion.

[0068] According to some embodiments of the present disclosure, a method for fabricating a memory device is provided. The method includes forming a bottom electrode layer over a substrate; forming a buffer layer over the bottom electrode layer; performing a surface treatment to a top surface of the buffer layer; depositing a resistance switch layer over the top surface of the buffer layer after performing the surface treatment; forming a top electrode over the resistance switch layer; and patterning the resistance switch layer into a resistance switch element below the top electrode.

[0069] According to some embodiments of the present disclosure, a method for fabricating a memory device is provided. The method includes forming a bottom electrode layer over a substrate; turning a top portion of the bottom electrode layer into a metal-containing oxide layer by introducing an oxide-containing gas to the top portion of the bottom electrode layer; depositing a resistance switch layer over the metal-containing oxide layer; forming a top electrode over the resistance switch layer; and patterning the resistance switch layer into a resistance switch element over the metal-containing oxide layer.

[0070] According to some embodiments of the present disclosure, a memory device includes a bottom electrode, a buffer element, a metal-containing oxide portion, a resistance switch element, and a top electrode. The buffer element is over the bottom electrode. The metal-containing oxide portion is over the buffer element, in which the metal-containing oxide portion has a same metal material as that of the buffer element. The resistance switch element is over the metal-containing oxide portion. The top electrode is over the resistance switch element.

[0071] The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may

make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

What is claimed is:

1. An integrated circuit (IC) structure, comprising:
an interconnect structure;
a first electrode over the interconnect structure;
a tantalum-containing layer over the first electrode, the tantalum-containing layer having oxide vacancies at a top surface of the tantalum-containing layer;
a high-k dielectric layer in contact with the top surface of the tantalum-containing layer; and
a second electrode over the high-k dielectric layer.
2. The IC structure of claim 1, further comprising:
a second electrode interposing the first electrode and the tantalum-containing layer.
3. The IC structure of claim 2, wherein a material of the second electrode is more inert to oxygen than a material of the first electrode.
4. The IC structure of claim 2, wherein the first electrode comprises a non-noble metal, and the second electrode comprises a noble metal.
5. The IC structure of claim 2, wherein the second electrode is formed of a material different from the tantalum-containing layer.
6. The IC structure of claim 1, wherein the tantalum-containing layer comprises a tantalum nitride sub-layer and a tantalum oxide sub-layer over the tantalum nitride sub-layer.
7. The IC structure of claim 6, wherein the high-k dielectric layer is in contact with the tantalum oxide sub-layer.
8. The IC structure of claim 6, wherein the tantalum oxide sub-layer is thinner than the high-k dielectric layer in a cross-sectional view.
9. The IC structure of claim 1, wherein the high-k dielectric layer has a width greater than a width of the second electrode in a cross-sectional view.
10. The IC structure of claim 1, further comprising:
spacers on opposite side surfaces of the second electrode.
11. The IC structure of claim 10, wherein the spacers are in contact with a top surface of the high-k dielectric layer.

12. An IC structure, comprising:
an interconnect structure;
a first electrode over the interconnect structure;
a metal nitride layer over the first electrode;
a metal oxide layer over the metal nitride layer;
a resistance switch element over the metal oxide layer;
and
a second electrode over the resistance switch element, the second electrode having a width smaller than a width of an interface formed by the metal nitride layer and the metal oxide layer in a cross-sectional view.
13. The IC structure of claim 12, wherein the metal nitride layer and the metal oxide layer comprise a same metal element.
14. The IC structure of claim 12, wherein the metal nitride layer is a tantalum nitride layer, and the metal oxide layer is a tantalum oxide layer.
15. The IC structure of claim 12, wherein the width of the second electrode is also smaller than a width of an interface formed by the metal oxide layer and the resistance switch element in the cross-sectional view.
16. The IC structure of claim 12, further comprising:
a second electrode disposed between the first electrode and the metal nitride layer, wherein the first electrode is a non-noble metal, and the second electrode is a noble metal.
17. An IC structure, comprising:
an interconnect structure;
a first electrode over the interconnect structure;
a ruthenium layer over the first electrode;
a tantalum nitride layer over the ruthenium layer;
a tantalum oxide layer over the tantalum nitride layer;
a high-k dielectric layer over the tantalum oxide layer;
and
a second electrode over the high-k dielectric layer.
18. The IC structure of claim 17, wherein the ruthenium layer is more inert to oxygen than a material of the first electrode.
19. The IC structure of claim 17, wherein the first electrode comprises a non-noble metal.
20. The IC structure of claim 17, wherein the tantalum oxide layer is in contact with a top surface of the tantalum nitride layer.

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